

CANalyst-II Analyzer (Top Version Pro)

Out-of-the-Box Test Manual

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I. Introduction

First of all, thank you for purchasing and using our products, we will always be happy to serve you!

This document is intended to guide first-time buyers of the CANalyst-II Analyzer (Top Version Pro) to test whether the CANalyst-II Analyzer (Top Version Pro) they receive works properly.

It can be ignored by experienced users or users who are more familiar with the operation of the USB-CAN bus adapter/CANalyst-II.

II. Test Steps

When you receive your CANalyst-II analyzer (Top Version Pro), please follow the steps described below to test it in order.

1. Install the Driver for the Hardware

Connect the CANalyst-II analyzer (Top Edition Pro) to one of the USB ports of your computer with the standard accessory "USB cable", and the computer will pop up the "Driver Installation Wizard".

Please refer to *"2.USB Driver Installation and Uninstallation Manual.pdf"* for detailed steps on driver installation.

2. Installation of Test Software

For detailed step-by-step instructions on the installation of the test software, please refer to *"3. USB-CAN Tool Debugging Software Installation and Usage Manual.pdf"*.

3. Connecting the Two Channels of the USB-CAN Bus Adapter

3.1 Short the Termination Resistor

Toggle R1 and R2 in the blue dip switches of the CAN1 channel on the left side of the CANalyst-II analyzer to the upper OFF position; and Toggle R1 and R2 in the red dip switches of the CAN2 channel on the right side to the lower ON position.

Note: Be sure to configure the termination resistor before connecting the device to the USB port of the computer via the USB cable, otherwise the test will not be timely and you need to plug and unplug the USB to retest.

3.2 Connect the CANH and CANL Signal Lines of CAN1 and CAN2 Channels.

Short the CANH of CAN1 channel and CANH of CAN2 channel, and short the CANL of CAN1 channel and CANL of CAN2 channel. The schematic diagram after shorting is shown below:

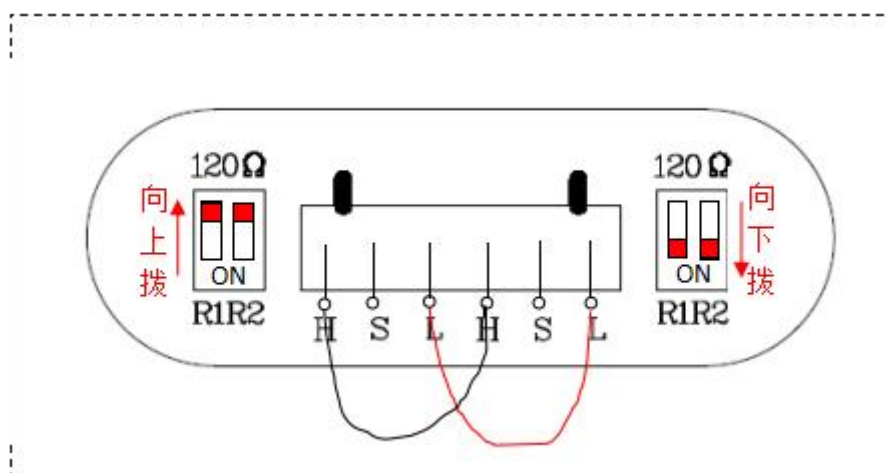


Figure 1 CANalyst-II Analyzer Test Wiring Diagram

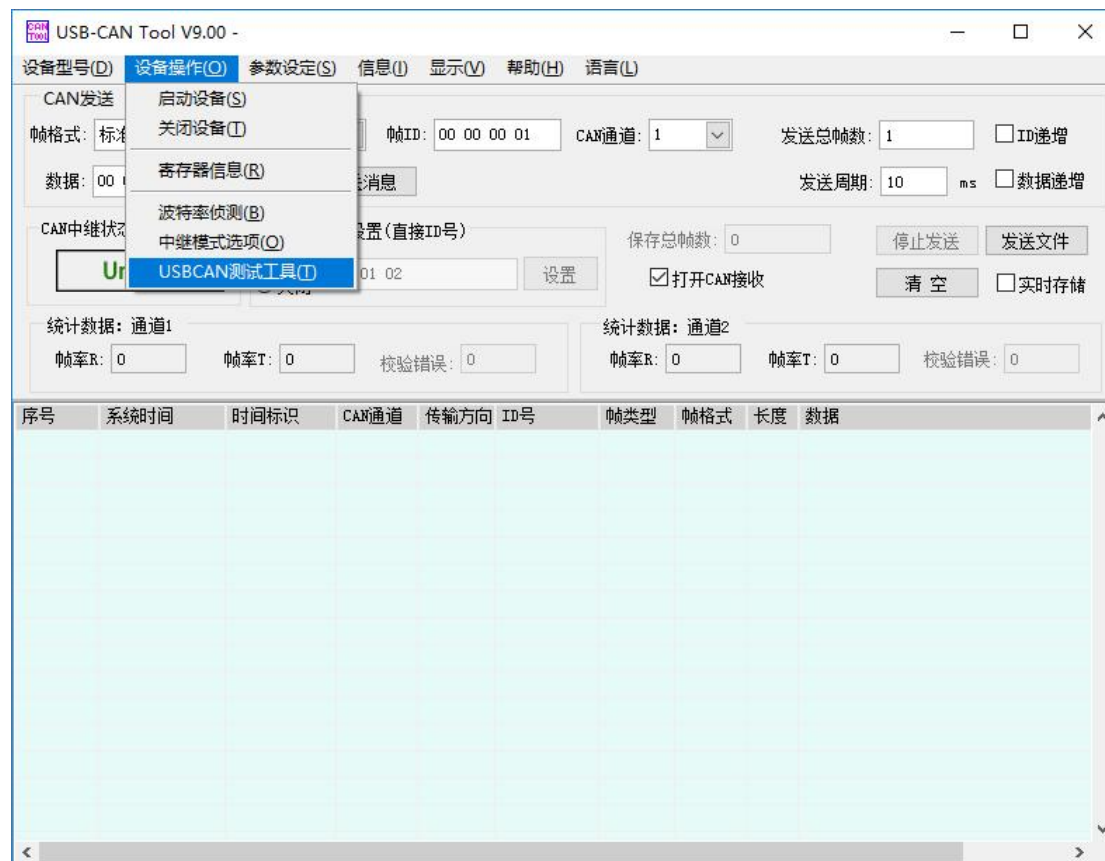
4. Run the USB-CAN Tool Test Tool

Run USB-CAN Tool debugging software, if the computer is installed in the protection software, such as 360 security guards, may open the exe program, a pop-up dialog box prompts the program to modify the registry entries, allow can be, which is LabVIEW run engine in the run of some of the program's operations:

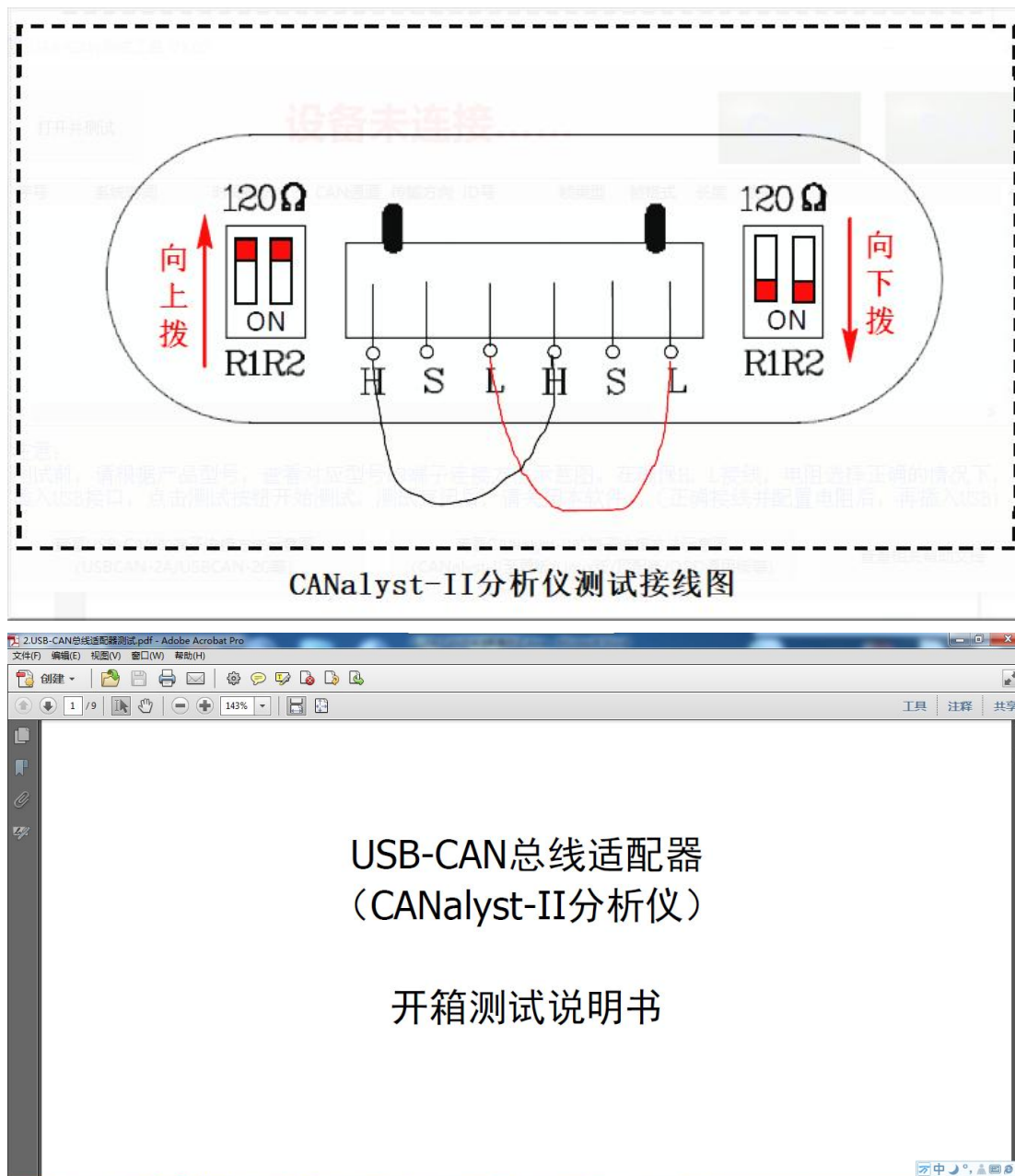


4.1 Automatic Testing With the USBCAN Test Tool

In the menu of the USB-CAN Tool interface, click on "Device Operation →USBCAN Test Tool":



Click on the three buttons below to view wiring diagrams, manuals for the corresponding model.



Connect the wires according to the diagram, plug in the device, click on the upper left corner of the "open and test" button, the software will automatically test (in order to open the device -> initialize CAN1/CAN2 two channels -> CAN1 sends a sequence, CAN2 to receive and verify -> CAN2 sends a sequence, CAN1 to receive and verify -> turn off the device -> the results are displayed):



If the following result appears, the test passes:

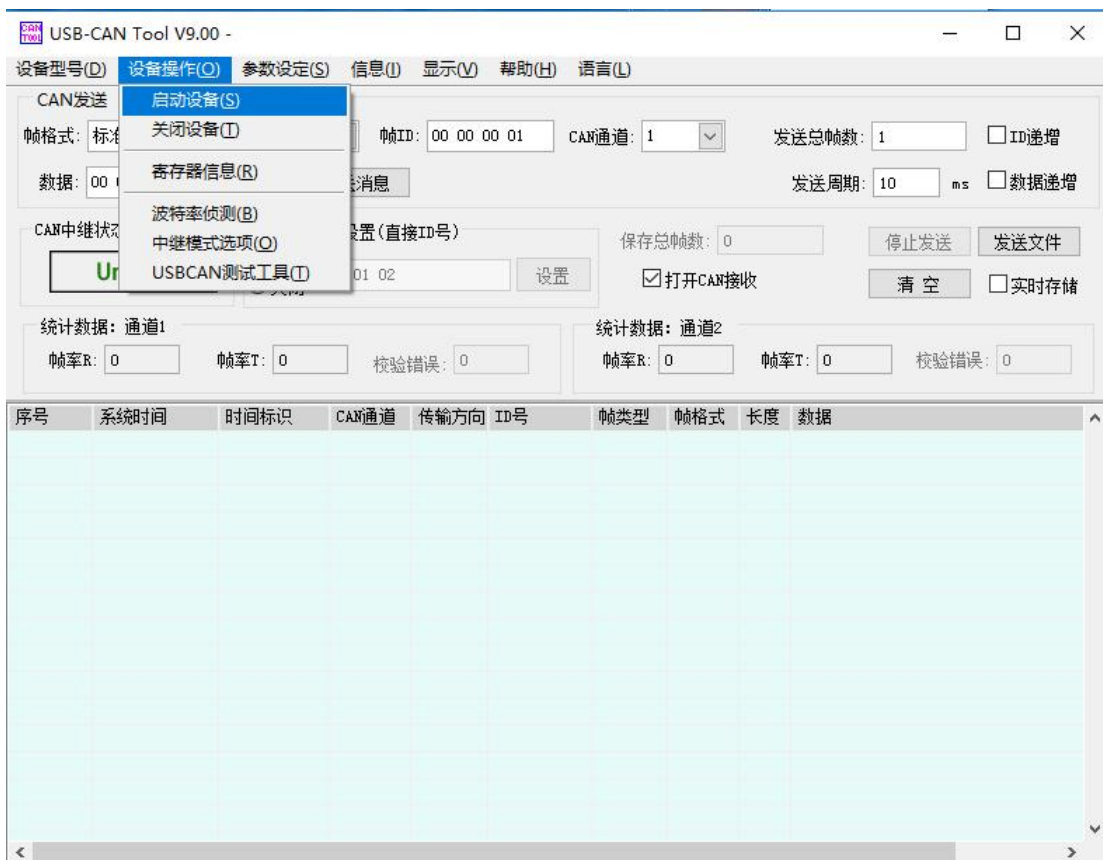


If the following results occur, follow the prompts to check wiring, terminating resistor configuration, and plug and unplug the USB to retest:



4.2 Manual Test

In the pop-up menu of USB-CAN Tool interface, click "Device Operation → to start the device:

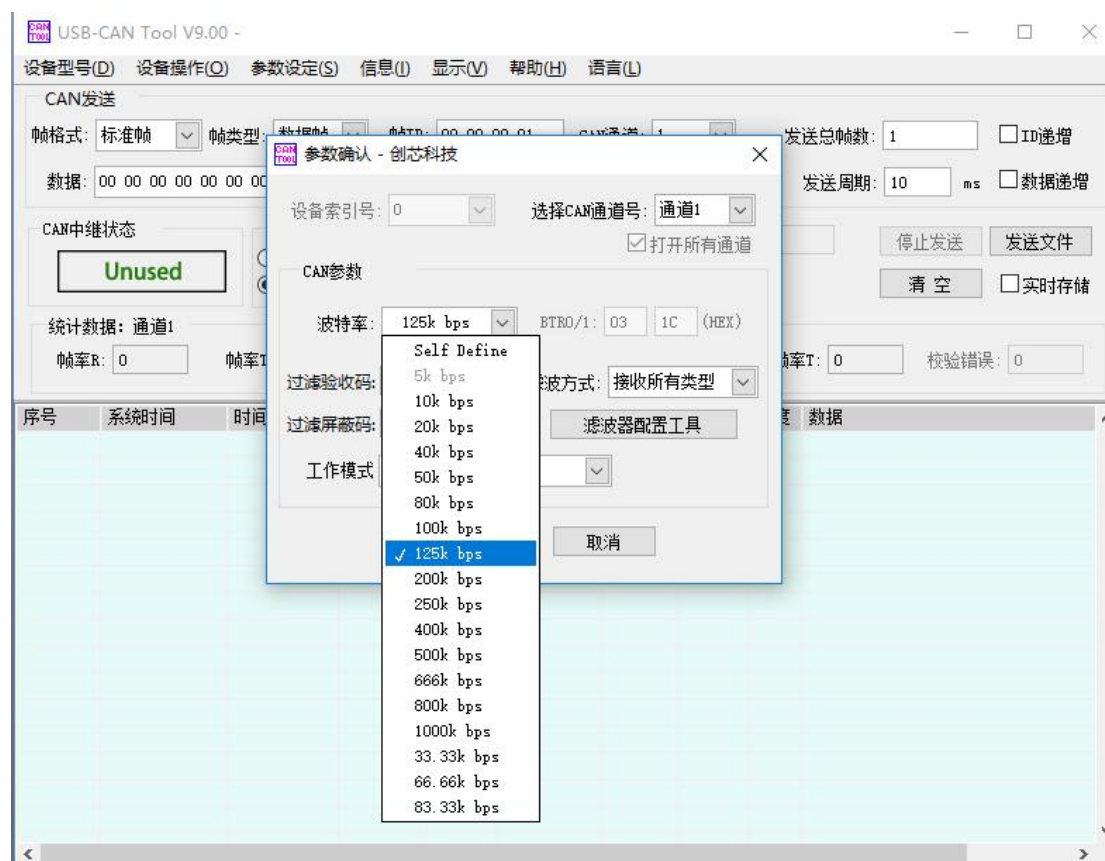


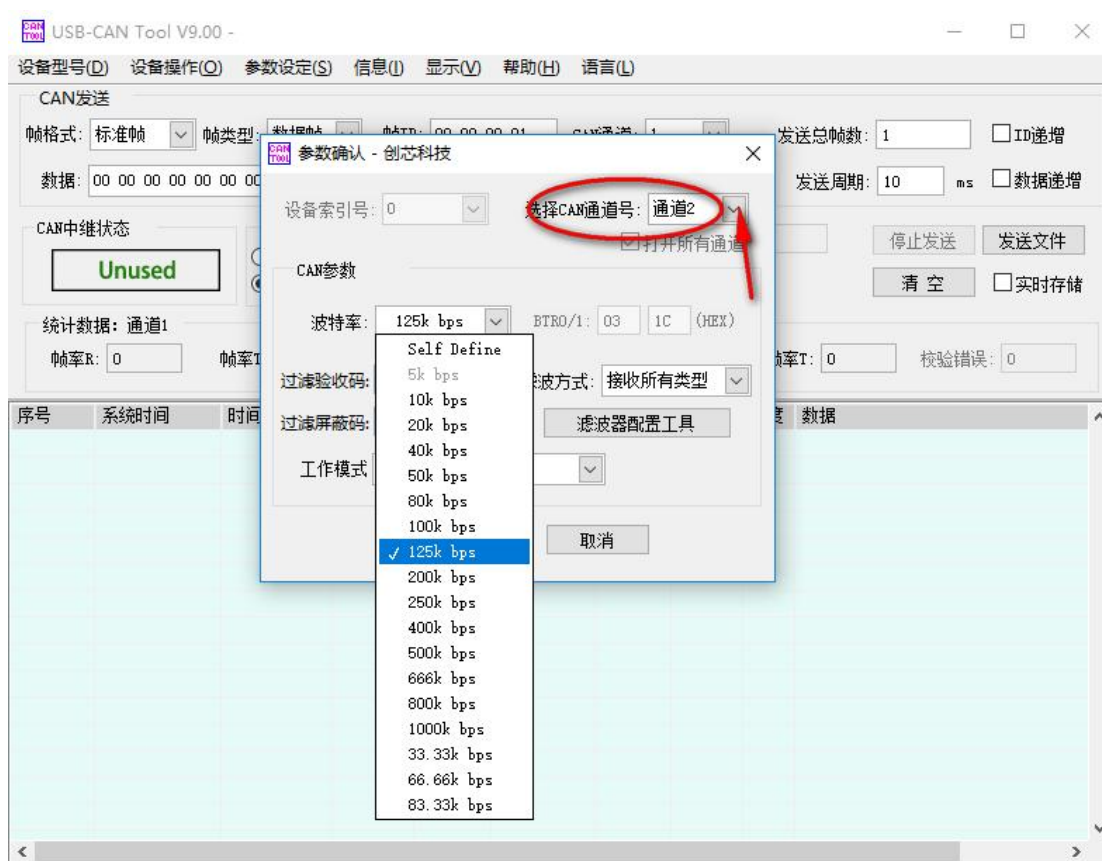
In the initialization parameter dialog box that pops up next, set the parameters of

CAN1 channel and CAN2 channel to the same baud rate, and configure the baud rate of CAN1 and CAN2 as 125K, and click "OK":

Note:

1. CAN2 channel baud rate need to switch the channel number to "channel 2" and then configure
2. When CAN2 channel is configured as high-speed CAN, CAN1 and CAN2 channels, the baud rate can be configured as 10K-1M, as long as the two channels are consistent.
3. When CAN2 channel is configured as low-speed fault-tolerant CAN, CAN2 channel can only configure the baud rate as 10K-125K, because CAN2 channel is fault-tolerant CAN bus baud rate up to 125K.

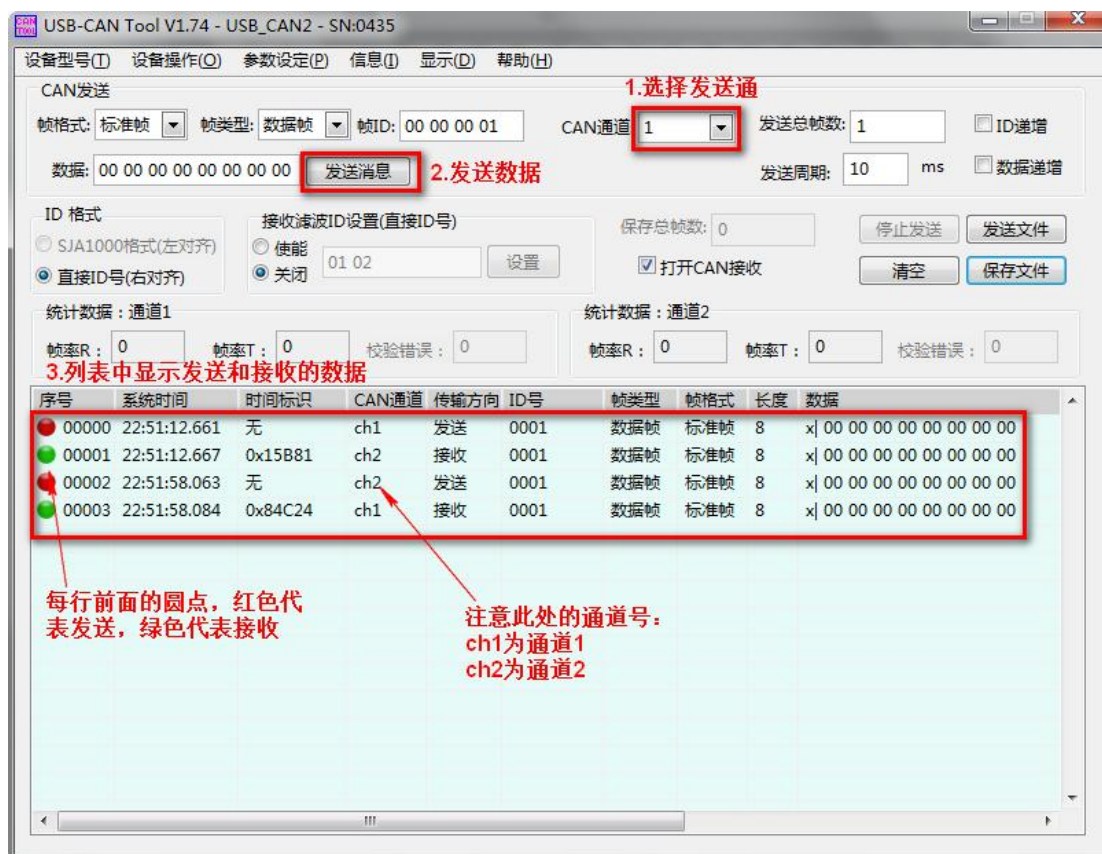




After opening successfully, select CAN1 channel on the interface, click "Send" button to send a frame of data, at this time, CAN2 channel will receive a frame of the same data, indicating that CAN1 channel sends normal, CAN2 channel receives normal.

Then switch the sending channel to CAN2 channel, click "Send" button to send a frame of data, at this time, CAN1 channel will receive a frame of the same data, indicating that CAN2 channel sends normally and CAN1 channel receives normally.

Note: If no data is received after clicking "Send", please check whether the terminal wiring is firm!



So far, CAN1 channel and CAN2 channel transmission and reception are normal, it means that the USB-CAN bus adapter is normal.

5. When the Test Is Normal By the Above Method, But It Can Not Send and Receive After Accessing the Bus, It Will Be Processed.

After the above steps are tested normally, it means that there is no problem inside the USB-CAN bus adapter, if it still can't receive or send data normally after connecting to the bus, most of them are caused by the physical wiring and baud rate mismatch, please further confirm the following sections:

5.1 Physical Wiring

Strictly follow the above steps for wiring.

Make sure the H and L of both channels are connected reliably.

Make sure the termination resistor is connected correctly.

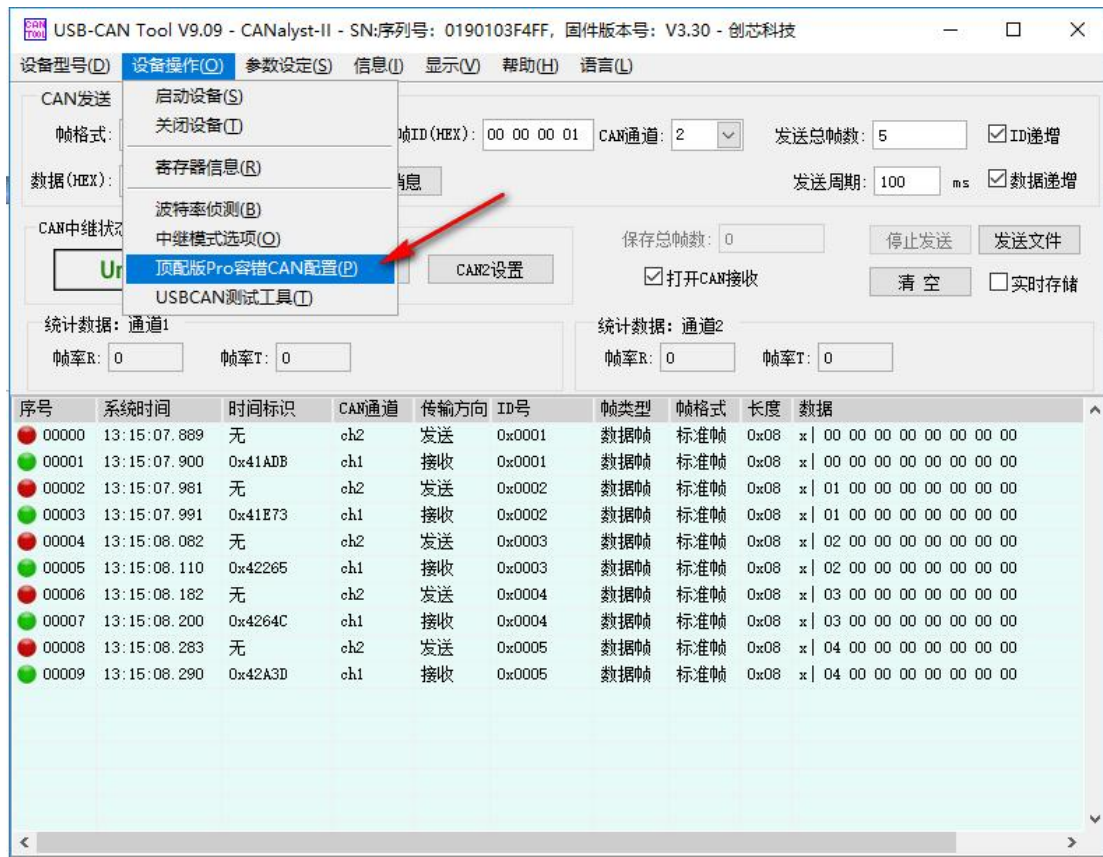
5.2 Wrong Parameter Matching

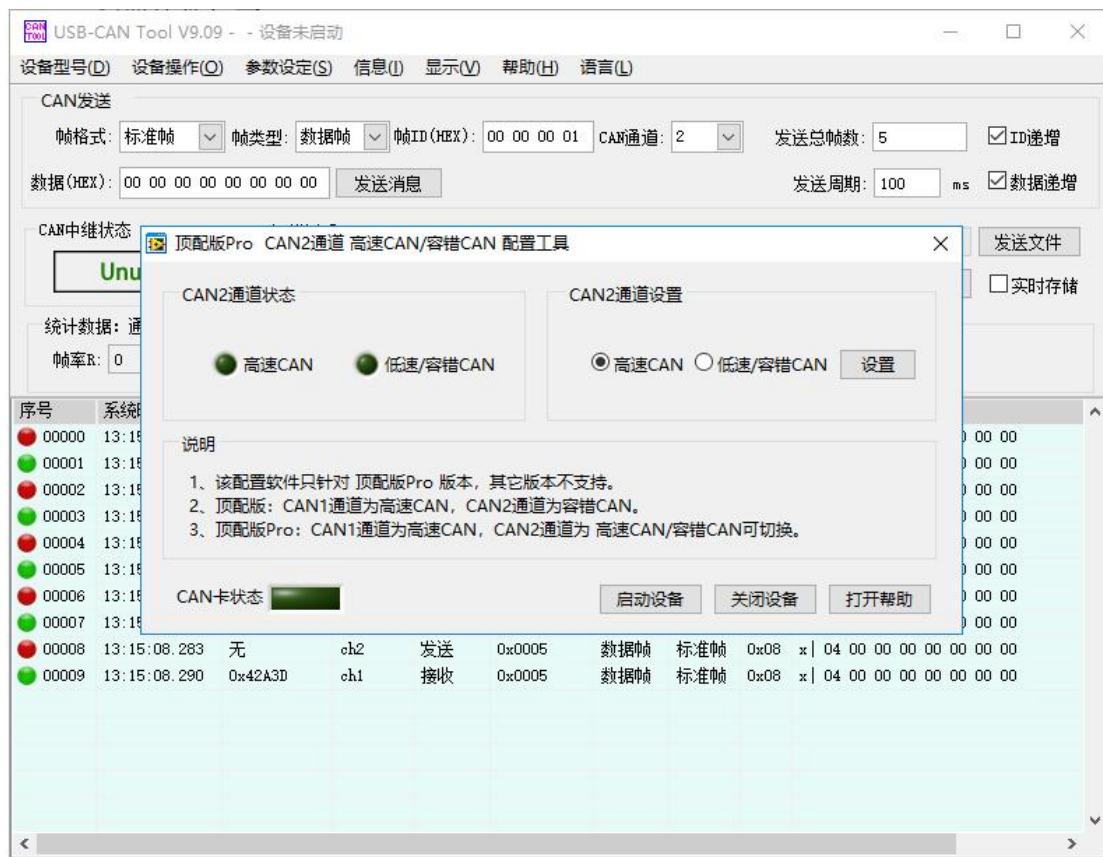
Strictly follow the above steps to configure the parameters, if it is used for the first time, use the default values directly. If the configuration has been modified, it is recommended to use the menu "Parameter Setting" -> "Restore Factory Configuration Parameters" and then operate according to the instruction manual.

When the baud rate of the target device is uncertain, you can refer to *"6.plugin 2: Baud Rate Detection Tool User Manual.pdf"* to detect the bus baud rate.

III. CAN2 Channel, Fault-Tolerant CAN Test

1. When the product is shipped from the factory, CAN2 channel is configured as high-speed CAN by default, and it needs to be configured when testing fault-tolerant CAN.





Click to start the device, and in the upper left corner of the plug-in interface, you can see the current status of the CAN2 channel: high-speed CAN.

On the right side, CAN2 channel setting area, choose to configure it as low-speed fault-tolerant CAN. After clicking on the setting, the system will restart automatically, and after restarting, it will take effect immediately. Just click to shut down the device. The configuration status will be saved until the next manual configuration.



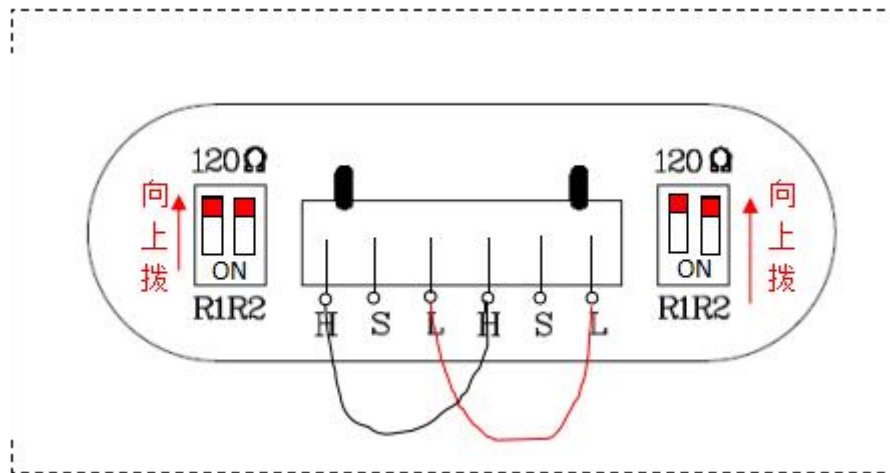


Figure 1 CANalyst-II Analyzer Test Wiring Diagram.

Dial the switches as shown above, here's what you should be aware of that all 4 switches need to be dialed to the top. At this time, the internal LED light, the display is solid green. If it is not solid green, you need to plug and unplug the USB.



Next, just test it with the test plugin.

